

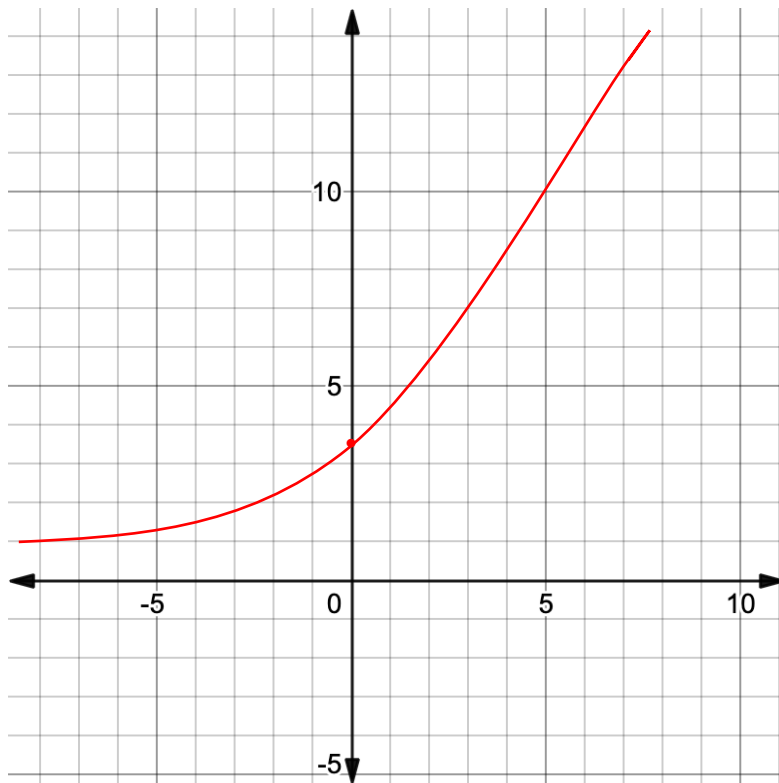
Algebra 1  
Unit 11  
Lesson 3 Practice

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

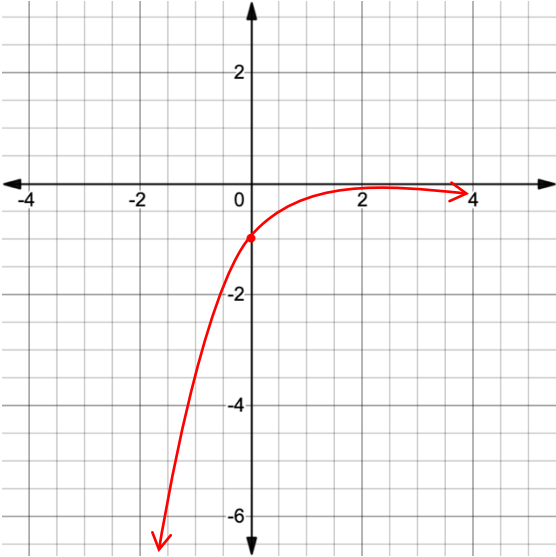
For questions 1-2, consider the function  $f(x) = \frac{7}{2}(1.23)^x$ .

1. Since  $a$  is ☒ positive  
☐ negative and  $b$  is ☐ between 0 and 1,  
☒ greater than 1, the function
- is ☒ increasing.  
☐ decreasing. The left end behavior of the graph is ☒  $y \rightarrow 0$   
☐  $y \rightarrow \infty$  and the
- right end behavior is ☐  $y \rightarrow 0$ .  
☒  $y \rightarrow \infty$ . The  $y$ -intercept of the function is ☐ 0.23.  
☐ 1.23.  
☒ 3.5.

2. Sketch the graph of  $f(x) = \frac{7}{2}(1.23)^x$ .



Graph the function of  $y = -(1 - 0.75)^x$ , complete the table of values, and then determine all key features.

| 3. Graph                                                                          | 4. Table of Values                                                                                                                                                                                                                                                             |     |     |    |     |    |    |   |    |   |                |   |                 |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|-----|----|----|---|----|---|----------------|---|-----------------|
|  | <table><tr><th><math>x</math></th><th><math>y</math></th></tr><tr><td>-2</td><td>-16</td></tr><tr><td>-1</td><td>-4</td></tr><tr><td>0</td><td>-1</td></tr><tr><td>1</td><td><math>-\frac{1}{4}</math></td></tr><tr><td>2</td><td><math>-\frac{1}{16}</math></td></tr></table> | $x$ | $y$ | -2 | -16 | -1 | -4 | 0 | -1 | 1 | $-\frac{1}{4}$ | 2 | $-\frac{1}{16}$ |
| $x$                                                                               | $y$                                                                                                                                                                                                                                                                            |     |     |    |     |    |    |   |    |   |                |   |                 |
| -2                                                                                | -16                                                                                                                                                                                                                                                                            |     |     |    |     |    |    |   |    |   |                |   |                 |
| -1                                                                                | -4                                                                                                                                                                                                                                                                             |     |     |    |     |    |    |   |    |   |                |   |                 |
| 0                                                                                 | -1                                                                                                                                                                                                                                                                             |     |     |    |     |    |    |   |    |   |                |   |                 |
| 1                                                                                 | $-\frac{1}{4}$                                                                                                                                                                                                                                                                 |     |     |    |     |    |    |   |    |   |                |   |                 |
| 2                                                                                 | $-\frac{1}{16}$                                                                                                                                                                                                                                                                |     |     |    |     |    |    |   |    |   |                |   |                 |

| Key Features                       |                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 5. Growth or Decay                 | decay since $b$ is .25 which is less than 1                                                                 |
| 6. Domain and Range                | Domain: $\mathbb{R}$ Range: $y < 0$                                                                         |
| 7. Increasing/Decreasing           | increasing<br>$-\infty < x < \infty$                                                                        |
| 8. Constant Percent Rate of Change | .25 = 25%                                                                                                   |
| 9. Left and Right End Behavior     | left: $x \rightarrow -\infty$<br>$y \rightarrow -\infty$ right: $x \rightarrow \infty$<br>$y \rightarrow 0$ |
| 10. Positive or Negative           | always negative $-\infty < x < \infty$                                                                      |